

DECLARATION OF CO-AUTHORSHIP

The declaration is for PhD students and must be completed for each conjointly authored article.

Please note that if a manuscript or published paper has eleven or less authors, all authors must sign a declaration of co-authorship. If it has twelve or more authors, only the PhD student, the corresponding author(s), the senior author, and the principal supervisor need to sign the declaration of co-authorship.

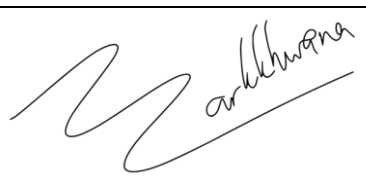
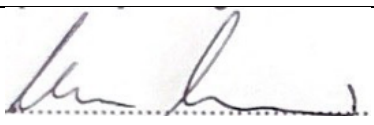
1. Declaration by	
Name of PhD student	Neil Alexandre Scheidwasser
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Name of principal supervisor	Samir Bhatt
Title of the PhD thesis	Pandemic preparedness in a vector: new representations of data from phylogenetic trees, protein structures, and animal tracking to aid preparedness

2. The declaration applies to the following article	
Title of article	Recombination of binding domains shapes structural evolution of <i>Betacoronavirus</i> spike proteins
Article status	
Published ☐	Accepted for publication ☐
Date:	Date:
Manuscript submitted ☐	Manuscript not submitted X
Date:	
If the article is published or accepted for publication, please state the name of journal, year, volume, page and DOI (if you have the information).	

3. The PhD student's contribution to the article <i>(please use the scale A-F as benchmark)</i>	
<u>Benchmark scale of the PhD-student's contribution to the article</u>	
A. Has essentially done all the work (> 90 %) B. Has done most of the work (60-90 %) C. Has contributed considerably (30-60 %) D. Has contributed (10-30 %) E. No or little contribution (<10 %) F. Not relevant	A, B, C, D, E, F
1. Formulation/identification of the scientific problem	A
2. Development of the key methods	B
3. Planning of the experiments and methodology design and development	B
4. Conducting the experimental work/clinical studies/data collection/obtaining access to data	A

3. The PhD student's contribution to the article (please use the scale A-F as benchmark) <u>Benchmark scale of the PhD-student's contribution to the article</u>		A, B, C, D, E, F
A. Has essentially done all the work (> 90 %) B. Has done most of the work (60-90 %) C. Has contributed considerably (30-60 %) D. Has contributed (10-30 %) E. No or little contribution (<10 %) F. Not relevant		
5. Conducting the analysis of data		
6. Interpretation of the results		
7. Writing of the first draft of the manuscript		
8. Finalisation of the manuscript and submission		A
Provide a short description of the PhD student's specific contribution to the article. ⁱ First and corresponding author. I identified the scientific problem, collected and curated the data, did the entire modelling and data analysis and wrote the first mock draft. I incorporated some feedback from co-authors on the methods used, on how to present the figures, interpretation of the results, and on shaping the first draft of the manuscript		

4. Material from another thesis / dissertationⁱⁱ	
Does the article contain work which has also formed part of another thesis, e.g. master's thesis, PhD thesis or doctoral dissertation (the PhD student's or another person's)?	Yes: ☐ No: X
If yes, please state name of the author and title of thesis / dissertation.	
If the article is part of another author's academic degree, please describe the PhD student's and the author's contributions to the article so that the individual contributions are clearly distinguishable from one another.	

5. Signatures of the authors incl. the PhD student				
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6. Signature of the principal supervisor

I solemnly declare that the information provided in this declaration is accurate to the best of my knowledge.

Date: 01.10.2025



Principal supervisor:

7. Signature of the PhD student

I solemnly declare that the information provided in this declaration is accurate to the best of my knowledge.

Date: 01.10.2025



PhD student:

Please learn more about responsible conduct of research on the [Faculty of Health and Medical Sciences' website](#).

ⁱ This can be supplemented with an additional letter if needed.

ⁱⁱ Please see Ministerial Order on the PhD Programme at the Universities and Certain Higher Artistic Educational Institutions (PhD Order) § 12 (4):

"Any articles included in the thesis may be written in cooperation with others, provided that each of the co-authors submits a written declaration stating the PhD student's or the author's contribution to the work."